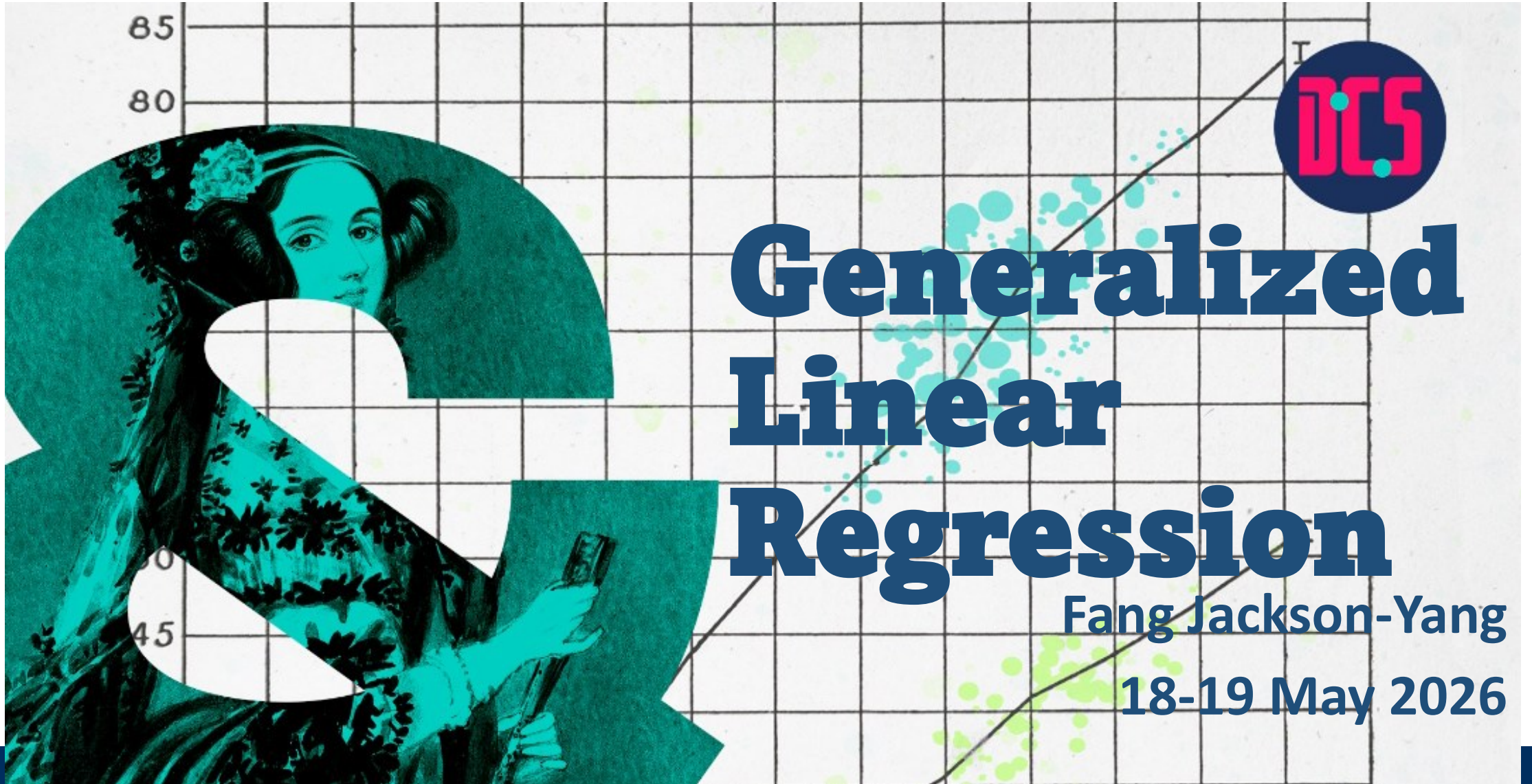
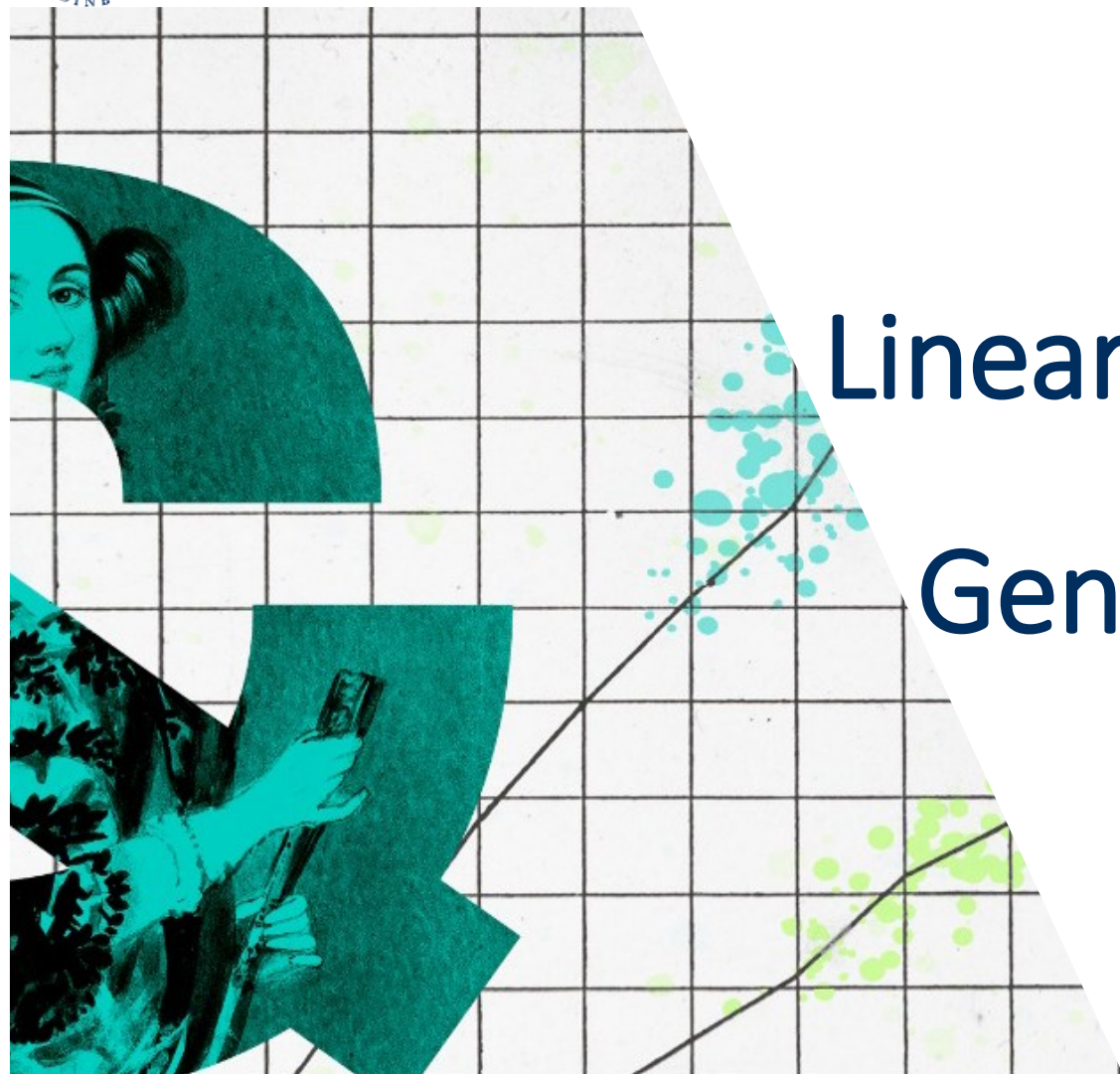




**Generalized
Linear
Regression**

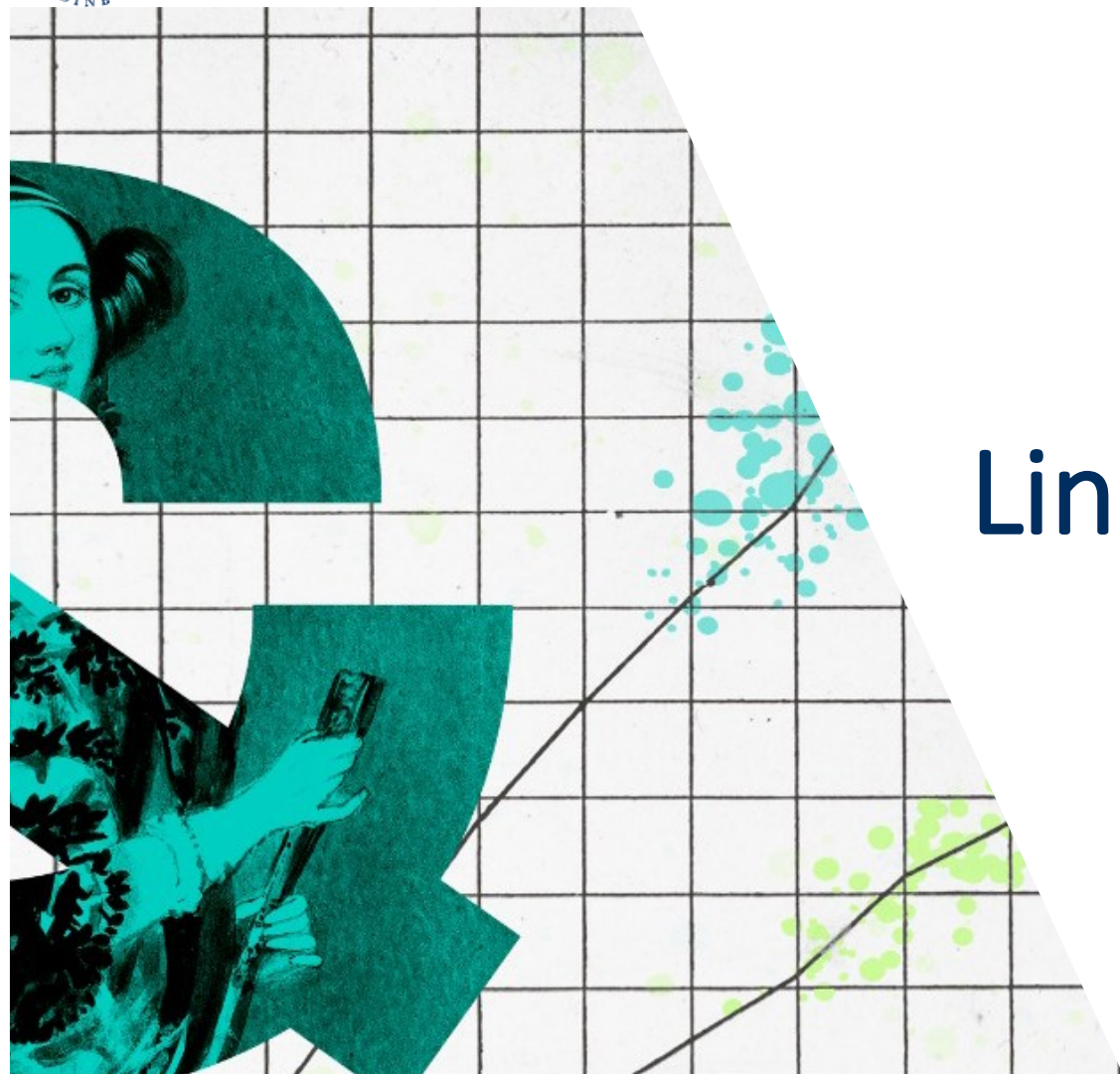
Fang Jackson-Yang

18-19 May 2026



Linear Model `lm()`

Generalized Linear Model `glm()`



Linear Model

`lm()`

Linear models - IGtime data

- Do undergraduates, FT_employee and retired people **differ** in the **amount of time** they spend on Instagram daily?

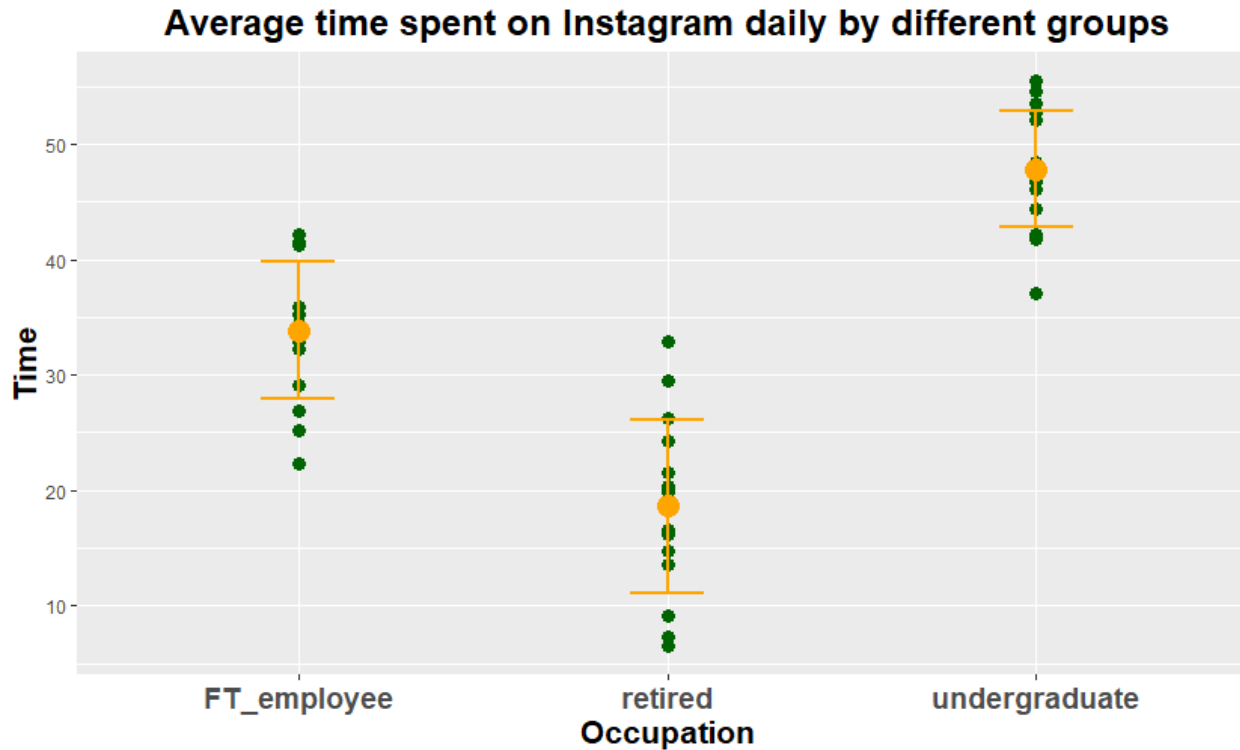
- continuous DV (outcome variable)

lm()



Linear models - IGtime data

`lm()`



- Plot the dataset using `ggplot()` in R

Linear models - lGtime data

lm()

Model Output

Call:
lm(formula = time_on_Instagram_daily ~ Occupation, data = data_lGtime)

Residuals:

Min	1Q	Median	3Q	Max
-12.1312	-3.5563	-0.1031	4.4437	14.2687

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	33.875	1.559	21.733	< 2e-16 ***
Occupationretired	-15.244	2.204	-6.915	1.36e-08 ***
Occupationundergraduate	13.956	2.204	6.331	1.00e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.235 on 45 degrees of freedom

Multiple R-squared: 0.796, Adjusted R-squared: 0.7869

F-statistic: 87.79 on 2 and 45 DF, p-value: 2.929e-16

estimate for
the mean of
the **baseline**
category

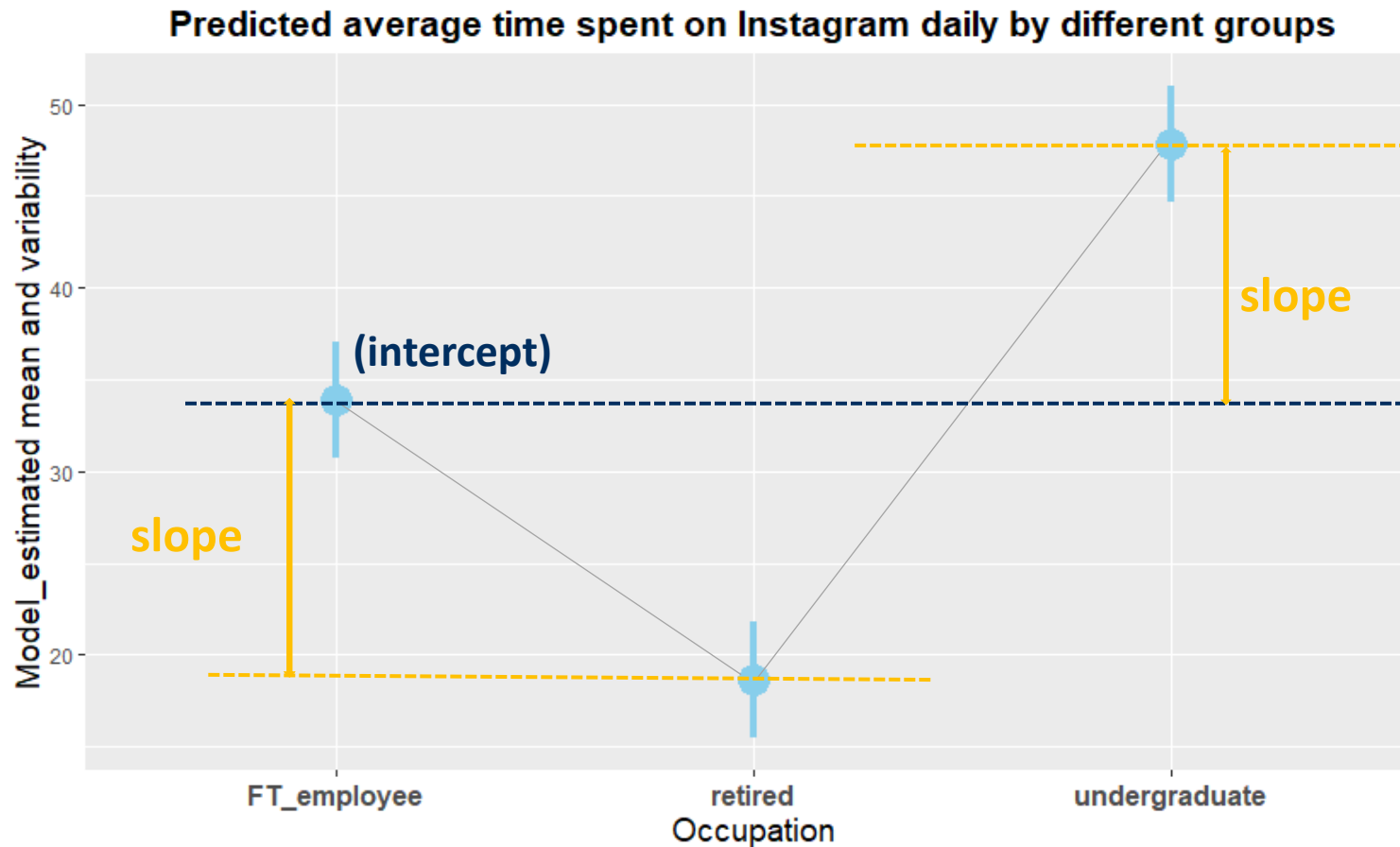
estimate for the
difference between
this category and the
baseline category

AKA **slope**

Linear models - lGtime data

`lm()`

- Interpret coefficients : intercept and slope



Linear models - IGtime data

lm()

- Interpret coefficients : intercept and slope

```
Call:
lm(formula = time_on_Instagram_daily ~ Occupation, data = data_IGtime)
```

Residuals:

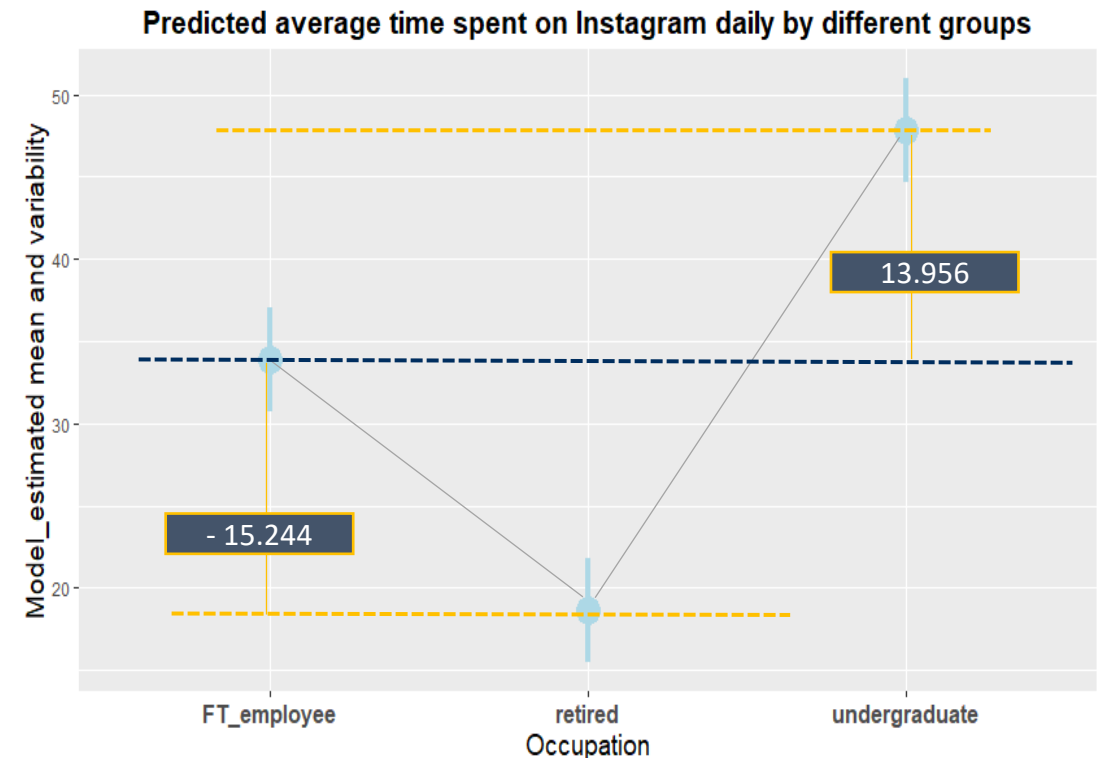
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Multiple R-squared: 0.796, Adjusted R-squared: 0.7869
F-statistic: 87.79 on 2 and 45 DF, p-value: 2.929e-16



What if the variable of our interest (i.e. the outcome variable) is categorical ?

➡ Generalised linear model **glm()**



Generalized Linear Model `glm()`

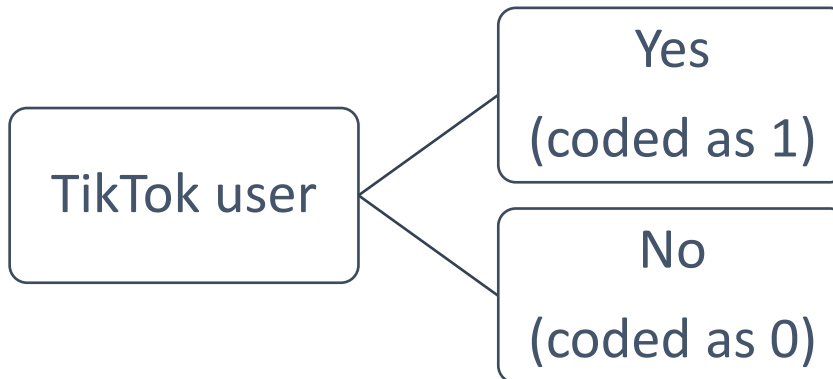


Generalised Linear models - IGtime data

- Would knowing how much time someone spends on Instagram daily help us explain the probability of them **being a TikTok user**?

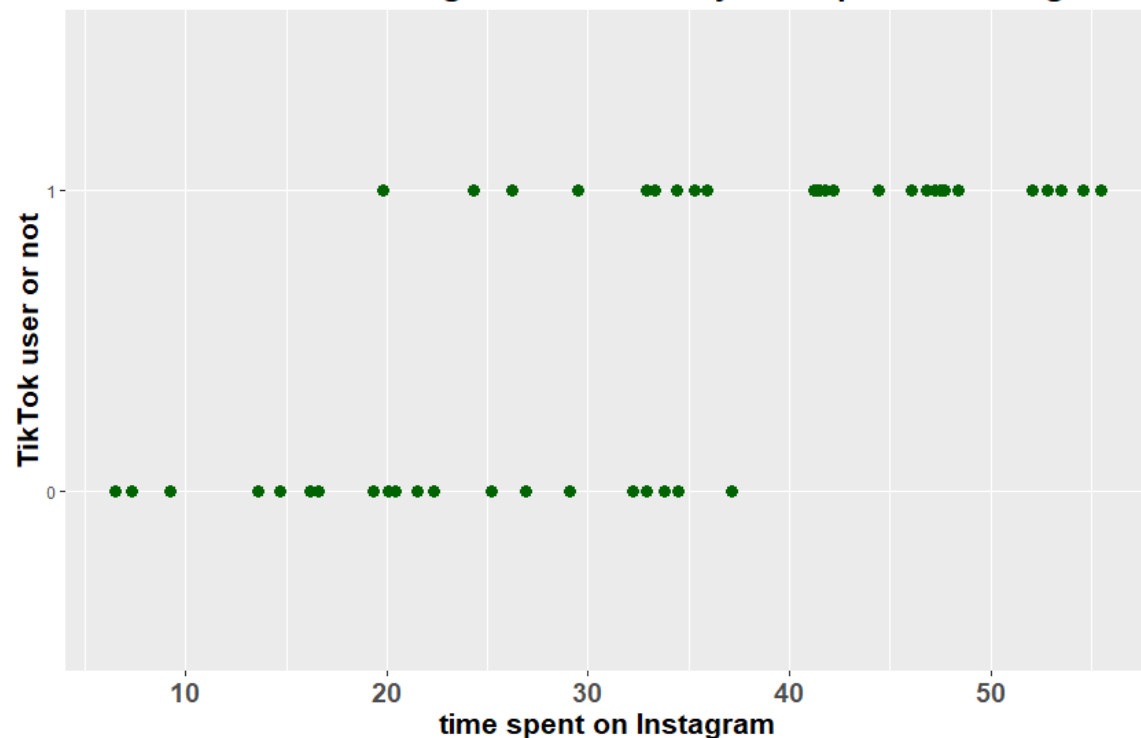
- categorical DV (outcome variable)

glm()



Generalized Linear Models - IGtime data

Relation between using TikTok and daily time spent on Instagram



glm()

- Plot the dataset using `ggplot()` in R

Generalised Linear models - IGtime data

Model Output in R

glm()

```
Call:
glm(formula = TikTok.user ~ time_on_Instagram_daily, family = "binomial",
    data = data_IGtime, na.action = na.exclude)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.7855	-0.4679	0.1532	0.4386	2.1040

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-6.06312	1.78521	-3.396	0.000683 ***
time_on_Instagram_daily	0.20027	0.05522	3.627	0.000287 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 65.203 on 47 degrees of freedom
Residual deviance: 32.868 on 46 degrees of freedom
AIC: 36.868

Number of Fisher Scoring iterations: 6

No p-value for
overall model

Generalised Linear models - IGtime data

glm()

Model Output in R

```
Call:
glm(formula = TikTok.user ~ time_on_Instagram_daily, family = "binomial",
     data = data_IGtime, na.action = na.exclude)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.7855  -0.4679   0.1532   0.4386   2.1040

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -6.06312    1.78521  -3.396  0.000683 ***
time_on_Instagram_daily  0.20027    0.05522   3.627  0.000287 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Analysis of Deviance Table

Model: binomial, link: logit

Response: TikTok.user

Terms added sequentially (first to last)

              Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                                47      65.203
time_on_Instagram_daily  1      32.335        46      32.868 1.298e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

No p-value for
overall model

Solution 1: run Likelihood Ratio Test

```
> anova(m_TikTok, test = "Chisq")
```


Generalised Linear models - IGtime data

glm()

Interpret coefficients

```
Call:
glm(formula = TikTok.user ~ time_on_Instagram_daily, family = "binomial",
     data = data_IGtime, na.action = na.exclude)
```

Deviance Residuals:

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Null deviance: 65.203 on 47 degrees of freedom
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AIC: 36.868

Number of Fisher Scoring iterations: 6

estimate for the
mean when the
predictor (IV) = 0

estimate for the
difference from 0
when the predictor
changes by 1 unit

AKA **slope**

Coefficients are in logit units (=log-odds)
Can be transformed to probabilities

Generalised Linear models - IGtime data

glm()

Interpret coefficients

```
Call:
glm(formula = TikTok.user ~ time_on_Instagram_daily, family = "binomial",
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```

Deviance Residuals:

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Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 65.203 on 47 degrees of freedom
Residual deviance: 32.868 on 46 degrees of freedom
AIC: 36.868

Number of Fisher Scoring iterations: 6

$$\text{Prob} = e^{-6.06} / (1 + e^{-6.06}) \\ = 0.2\%$$

$$\text{Prob} = e^{(-6.06+0.2)} / (1 + e^{(-6.06+0.2)}) \\ = 0.3\%$$

Coefficients are in logit units (=log-odds)

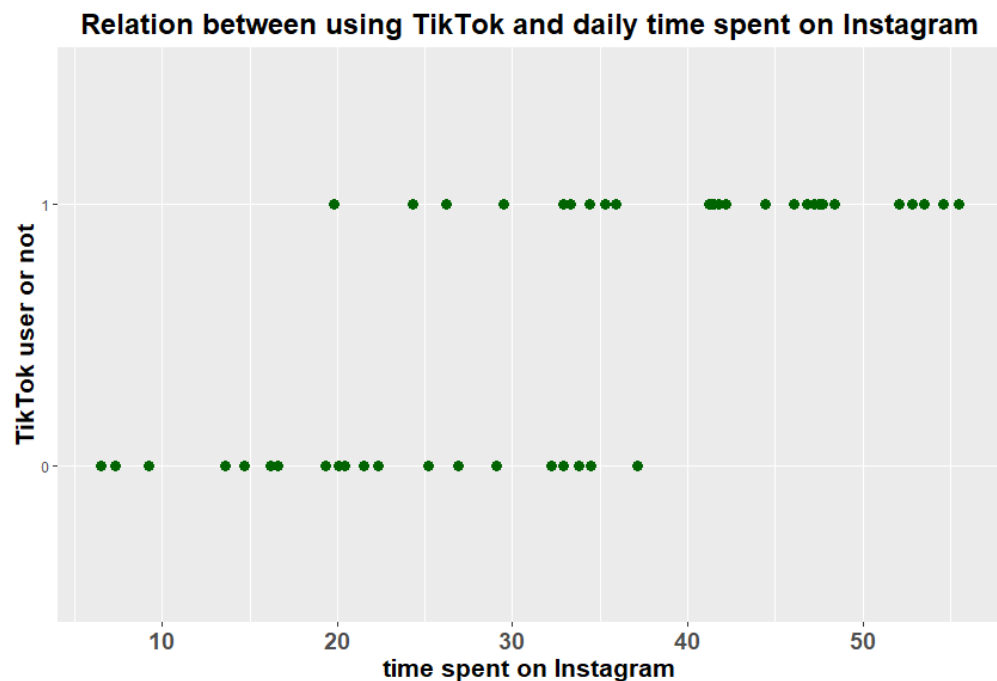
Can be transformed to probabilities: $\text{prob}(x) = \text{odds}(x) / (1 + \text{odds}(x))$

```
> exp(coef(m_TikTok)[1] + x * coef(m_TikTok)[2])
```

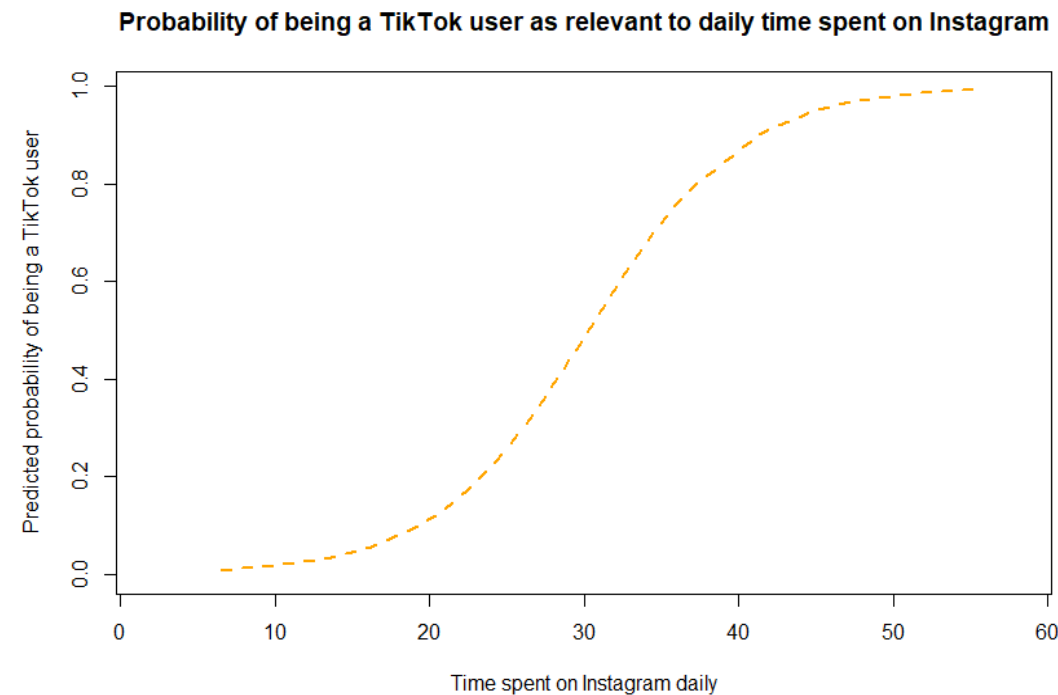
Generalized Linear Models - IGtime data

glm()

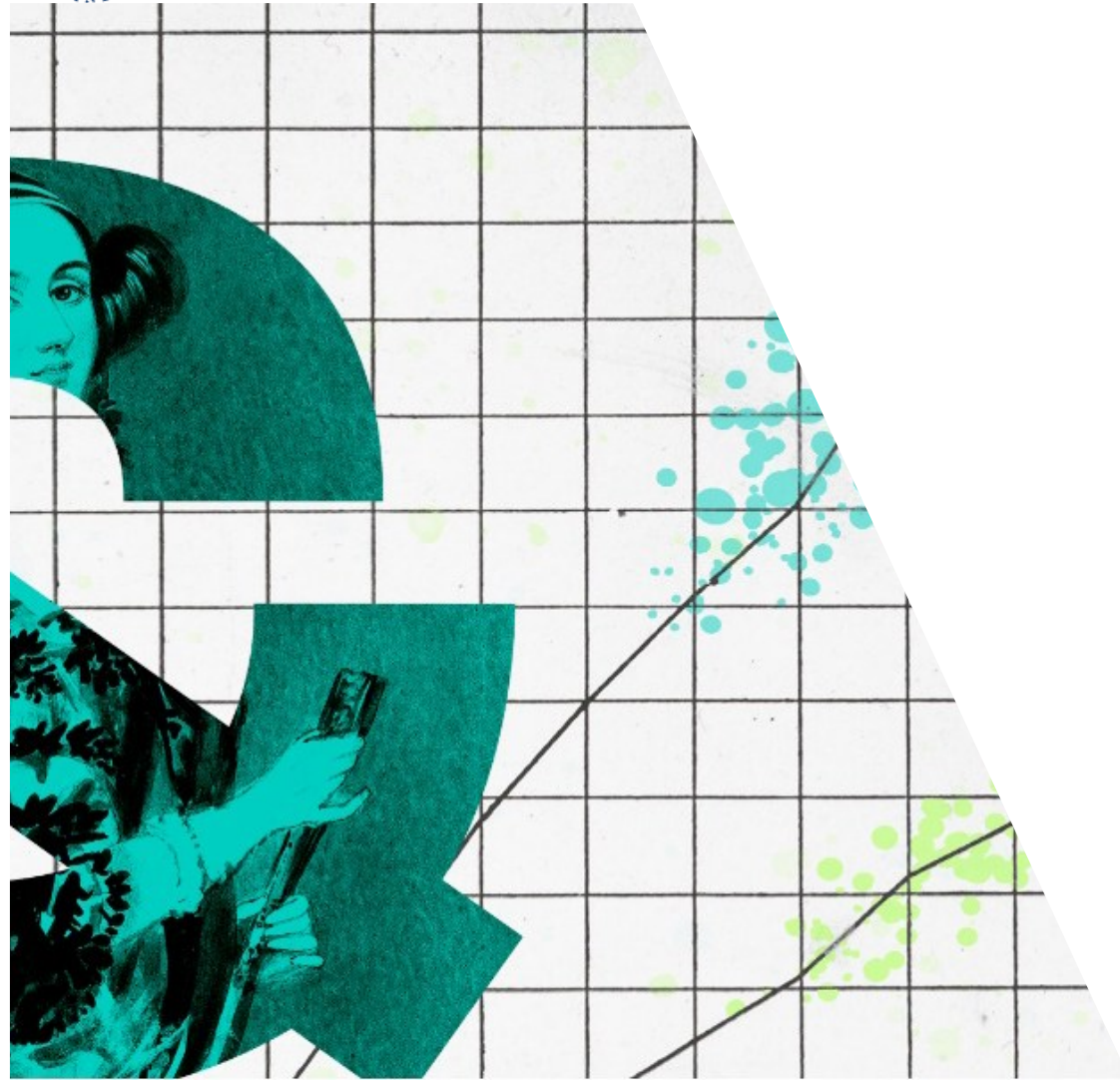
- Data



- Model (glm)



Model Assumption Checks



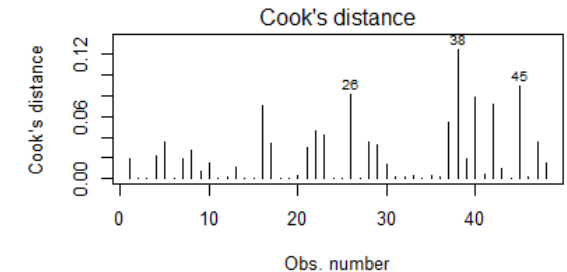
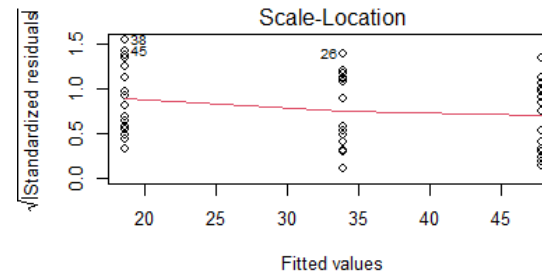
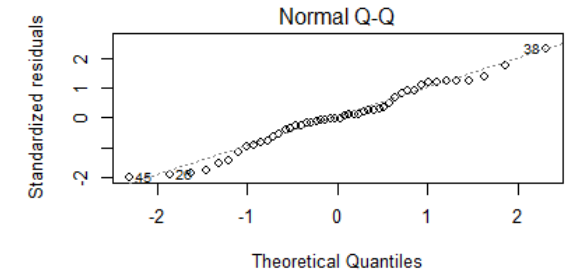
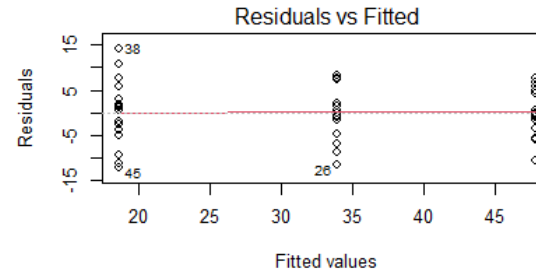
LINEAR MODEL ASSUMPTION CHECKS

- Required(LINE):

- Linearity of relationships
- Independence of residuals
- Normality of residuals
- Equal variances for residuals

- Desirable:

- uncorrelated predictors
(no collinearity) for multiple regression
- no “bad” (overly influential) observations



`lm()`

GENERALIZED LINEAR MODEL ASSUMPTION CHECKS

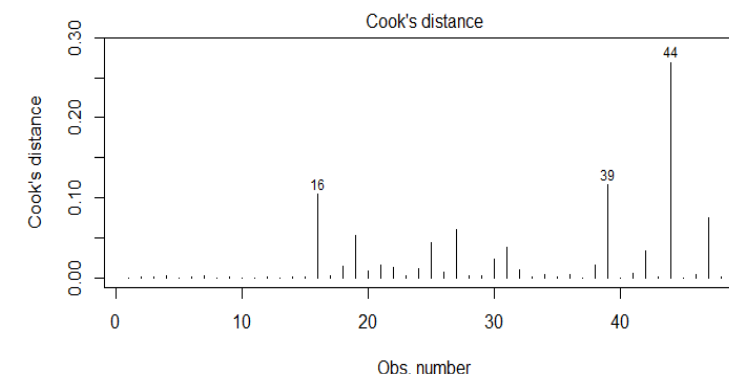
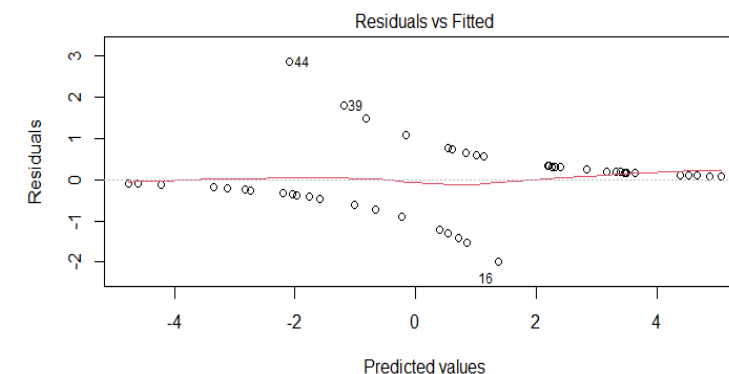
- Required(LINE):

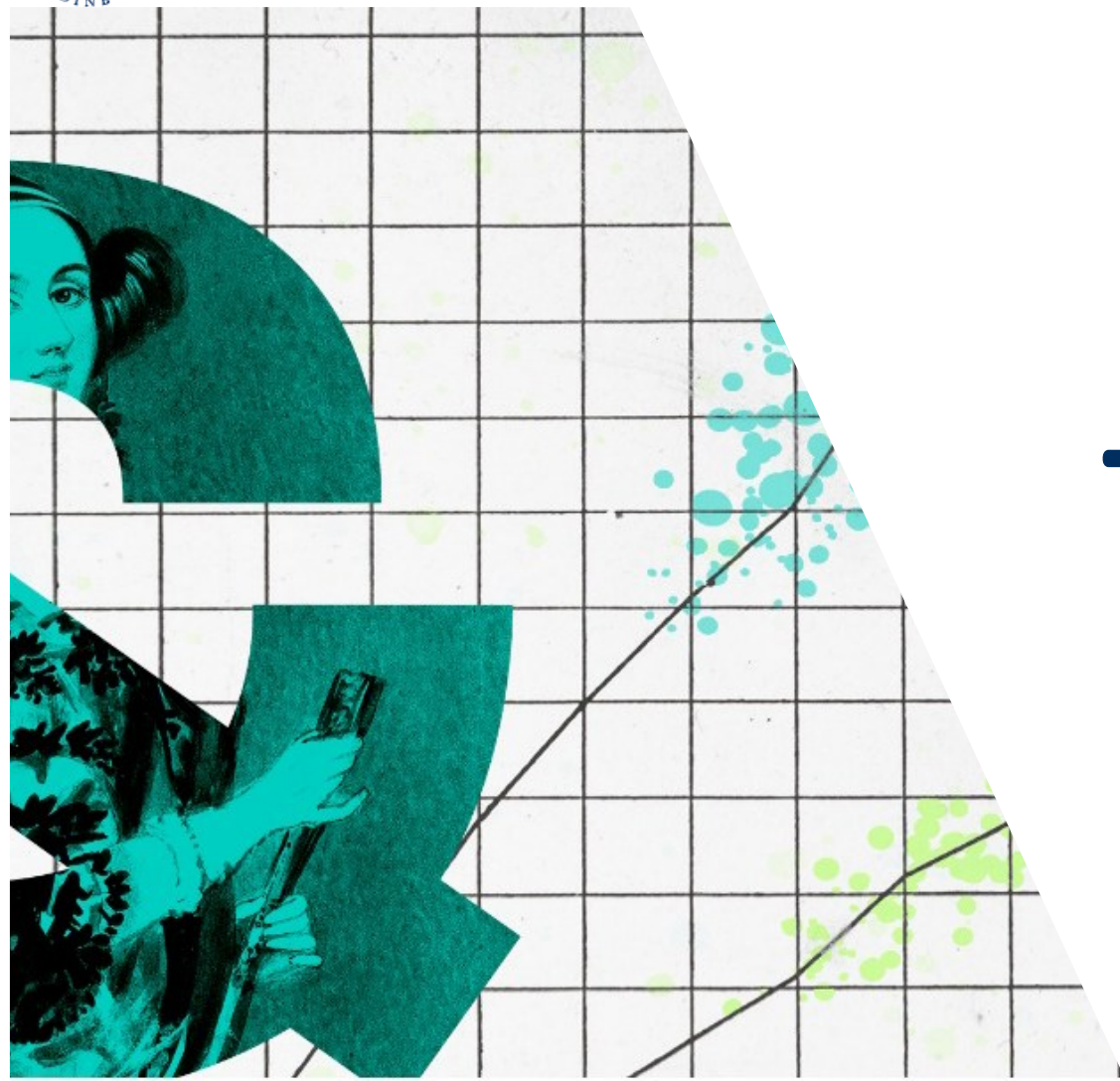
- Linearity of relationships
- Independence of residuals
- ~~- Normality of residuals~~
- ~~- Equal variances for residuals~~

`glm()`

- Desirable:

- uncorrelated predictors
(no collinearity) for multiple regression
- no “bad” (overly influential) observations
- large samples (due to maximum likelihood fitting)





THANK YOU

